Video & Pixel Manipulation

p5js

Motivation

The challenges is to explore p5js as integrated tool of video production.

How might that work?
What are the obstacles, points of friction?

Code Process

01. Research

02. Tutorials on repreat

03. Document and manipulate

03. Write from scratch

get()

loadPixels()

01. Pixel Manipulation

01. Pixels Arrays

02. Pixel Density

03. Functions to access array

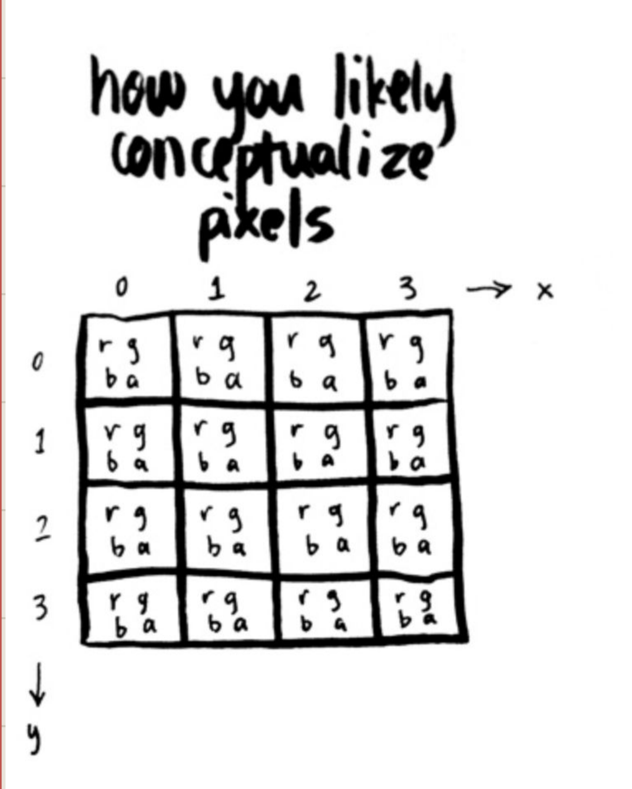
04. Updating Pixels

loadPixels()

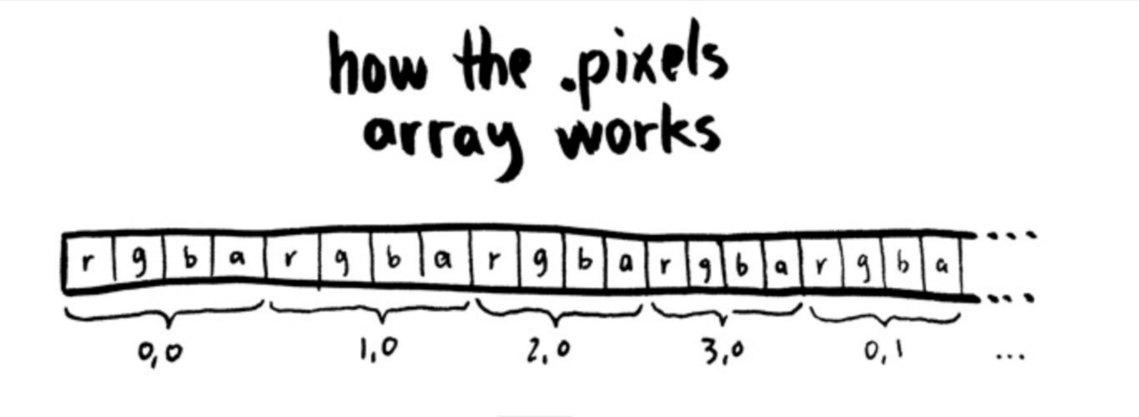
get()

updatePixels()

Pixel Arrays



Understanding How Pixel Arrays Work



.get

Gets the x and y position of each pixel.

Parameters

x	Number: x-coordinate of the pixel.
y	Number: y-coordinate of the pixel.
w	Number: width of the subsection to be returned.
h	Number: height of the subsection to be returned.

2D Array

loadPixels

Loads the current value of each pixel on the canvas into the pixels array.

1D Array, stores RGBA

.get

2D Array

	x = 0	x = 1	x = 2	x = 3
y = 0	0	1	2	3
y = 1	4	5	6	7
y = 2	8	9	10	11
y = 3	12	13	14	15

loadPixels

1D Array, stores RGBA

	x = 0	x = 1	x = 2	x = 3
y = 0	0, 1, 2, 3	4, 5, 6, 7	8, 9, 10, 11	12, 13, 14, 15
y = 1	16, 17, 18, 19	20, 21, 22, 23	24, 25, 26, 27	28, 29, 30, 31
y = 2	32, 33, 34, 35	36, 37, 38, 39	40, 41, 42, 43	44, 45, 46, 47
y = 3	48, 49, 50, 51	52, 53, 54, 55	56, 57, 58, 59	60, 61, 62, 63

Looping thru

```
let index = ((j*vid.width) +i) *4;
```

Convert **1D array** to **Index** in order to access **RGBA values**

```
// Red.
pixels[index] = 0;
// Green.
pixels[index + 1] = 0;
// Blue.
pixels[index + 2] = 0;
// Alpha.
pixels[index + 3] = 255;
```

Width * Height * 4

(elements of RGBA)

Pixels are greater

Math is Important

```
    //we, we must use. This grabs the pixel data
from the camera, which is larger than the canvas.
'
    let offset = ((cy*vid.width)+cx)*4;
```

Limitations and Friction

Production & Post

Obstacles

P5js fails to recognize black magic pocket camera input

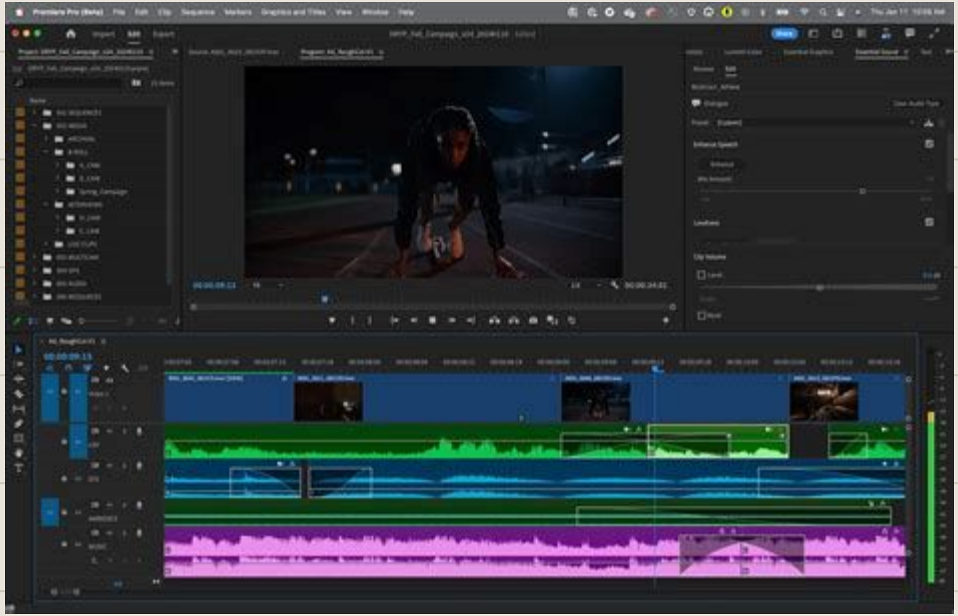
Production

Obstacles

Post Production

Work arounds are required to export video

1. Screen recordings
2. Stills



Video Manipulation

A series of video clips showcasing the abilities p5js pixel manipulation